

Dataset Name: Indian National Family Health Survey (NFHS-5) District Factsheet Data

Source: Ministry of Health and Family Welfare, Government of India (Data at District Level)

The dashboard focuses on simplicity and high policy impact:

- **Maternal & Child Survival Tracker:** Identify districts with concurrent low rates of **Institutional Births** and **Full Immunization**.
- **Education-to-Health Dividend Analysis:** Quantify the inverse correlation between **Women's Higher Education** (≥ 10 yrs) and **Child Marriage** rates.
- **Family Planning Intervention Priority:** Isolate districts based on a high **Total Unmet Need** despite low **Modern Contraceptive Use**.
- **Nutrition & Risk Assessment:** Map the geographic burden of **Child Stunting** alongside **Adult Overweight/Obesity** (Dual Burden).
- **Healthcare Financial Equity:** Benchmark districts by **Average Out-of-Pocket Expenditure (OOPE)** per public delivery to identify cost barriers.

India's vast public health landscape requires granular, district-level data for effective resource allocation. The problem is that national or state-level averages mask significant local failures. This dashboard's purpose is to move beyond mere reporting and use **correlation analysis** (e.g., Education vs. Health) and **quadrant analysis** (e.g., Unmet Need vs. Use) to pinpoint **specific geographical hotspots** where policy intervention will yield the maximum return.

- **Tool:** Microsoft Power BI (Power Query and DAX)
- **Data Cleaning:**
- **Text Cleanup:** Simplified overly descriptive column headers and trimmed whitespace from **District Names** and **State/UT**.
- **Handling Non-Numerics:** Replaced placeholders (*, -) used for suppressed data with **Null** in Power Query.
- **Data Typing:** Converted all percentage and value columns (including **OOPE**) from Text to **Decimal Number**.
- **Transformations (DAX):** Created necessary measures such as the **Dual Burden Index** (average of stunting and obesity) and calculated **Private Institutional Births** (Total - Public).
- **Education is the Key Intervention:** The dashboard strongly confirms the "**Education Dividend**," showing that investment in women's secondary education is one of the most effective tools for lowering social pathologies like child marriage.

Service Delivery Failure: A significant gap exists between high **Institutional Births** and low **Full Immunization**, indicating a failure in **post-maternity follow-up care** rather than facility access itself.

Financial Barrier: Specific districts show **disproportionately high OOPE** for public facility deliveries, undermining the intended equity and accessibility of government healthcare.

Focus Area	Strategic Recommendation	Actionable Implication
Program Integration	Mandate the convergence of MCH and Immunization programs using a unified KPI to ensure complete care.	Health Department: Restructure reporting to track children from birth to full vaccination status.
Resource Allocation	Prioritize OOPE Outliers: Launch immediate audits and regulatory checks on districts with the highest public healthcare costs.	Financial Wing: Establish a formal system to flag and investigate high-cost public facilities.
Social Policy	Fund Education as Health: Direct convergence funds toward increasing girls' secondary school enrollment in districts with the highest Child Marriage rates.	State Planning/Social Welfare: Use the dashboard to justify targeted infrastructure spending.

Team: Data Crafters

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